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PARKING PAWL SAFETY LATCH SYSTEM

Abstract

The present disclosure relates to a system and method to stop precession of a flywheel enclosure used in a gyroscopic roll stabilizer in aquatic vessels. The system and method includes the use of a pawl safety latch capable of interlocking connection with a gimbal shaft to prevent rotation of the gimbal shaft and prevent precession of the corresponding flywheel enclosure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit and priority to U.S. provisional application Ser. No. 63/557,862 filed on Feb. 26, 2024, the foregoing hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a system and method to stop precession of a flywheel used in a gyroscopic roll stabilizer in aquatic vessels in the event of braking equipment failure.

BACKGROUND

[0003] As an aquatic vessel sits in the water, it is moved by waves, currents, and tides. These forces cause an aquatic vessel, such as a boat, to move through six degrees of freedom: three rotational movements, referred to as roll (tilting of a vessel about its longitudinal axis), pitch (up/down rotation along the transverse axis), and yaw (rotation about the vessel's vertical axis), and three translational movements, referred to as surge (linear front to back motion), sway (linear side to side motion), and heave (linear vertical motion). These motions can make life on an aquatic vessel uncomfortable or dangerous.

[0004] Historically, these motions were counteracted by stabilizing ballast, such as weighed kegs, rocks, water tank, or other heavy matter placed in the hull of the vessel to lessen the effects of these motions. These historical methods to combat unwanted vessel motion, and in particular, antiroll effects, were heavy, slowed the vessel's speed, and occupied the cargo space, making such space unusable. Modern antiroll technology has improved on these historical antiroll means. One antiroll technology is gyroscopic stabilization. Gyroscopic stabilization involves a flywheel disposed in a neutral orientation wherein the axis of rotation of the flywheel is generally parallel to either the vertical or horizontal axis of a vessel. The flywheel rotates within a flywheel enclosure producing torque and angular momentum. The flywheel enclosure is attached to a frame that is rigidly mounted to the vessel and is able to rotate about a gimbal axis that is perpendicular to the axis of flywheel rotation. As the vessel rolls, the flywheel enclosure precesses about the gimbal axis and generates torque about the roll axis of the vessel. This torque counteracts the roll of the vessel generated by a body of water, thereby stabilizing the vessel.

[0005] Control of the precession rate of the flywheel enclosure is required to realize vessel stabilization in aquatic applications. The precession rate of the flywheel enclosure is controlled through a precession braking system, which can be comprised of a combination of hydraulic, mechanical, and/or electrical components. Failure of one or more of these components could result in an uncontrolled precession rate, presenting compromised performance and safety hazards.

[0006] Therefore, there exists a long-felt, unmet need for a safer, more effective system and method to safely and securely stop gyroscopic precession in the event of an electrical or mechanical failure in aquatic vessels, and more particularly failure of any components of the precession braking system.

SUMMARY

[0007] The present disclosure relates to a system and method to utilize a pawl to engage with a gimbal shaft from a gyroscopic stabilizer system to halt precession in the event of a power failure (or failure of braking equipment) on an aquatic vessel.

[0008] In one embodiment of the system and method, an assembly to stop precession of a flywheel enclosure in the event of a failure of braking equipment is disclosed and may include: a pawl with a pawl front latching pad, a pawl back latching pad, a pawl end stop, a pawl pivot bushing, a pawl compliance shaft, a pawl compliance sleeve, a solenoid and solenoid plunger, and a gimbal shaft end with apertures for the pawl, wherein the gimbal shaft is connected to a brake system, wherein the pawl is connected to the solenoid plunger, and wherein the solenoid plunger is electronically coupled to the solenoid such that in the event of a power failure or other braking failure, the solenoid can engage the pawl to interact with the apertures on the gimbal shaft end.

[0009] The braking equipment used to stop the precession of a flywheel enclosure in a gyroscopic roll stabilizer is commonly made of hydraulic, mechanical, magnetic, or electrical components, or a combination thereof. If the braking equipment fails to operate as intended, the effectiveness of the braking equipment may be compromised and produce a safety hazard that is only mitigated through locking the flywheel enclosure from precessing. The parking pawl safety latch is a novel concept to stop precession of the flywheel enclosure in the event of a failure of the precession braking equipment.

[0010] In this preferred embodiment of the present disclosure, the parking pawl safety latch stops uncontrolled precession of the flywheel enclosure without the use of the compromised precession braking system.

[0011] In another embodiment, the parking pawl safety latch assembly may include a gimbal shaft with two openings which act as lock points and parking pawl safety latch that may include a pawl disposed adjacent a distal end of a latch that is shaped to engage with one of the openings to immediately lock the gimbal shaft in place to prevent any further rotation of the flywheel enclosure and prevent disengagement of the pawl from the opening.

[0012] The parking pawl safety latch assembly of the present disclosure can be used to stop rotation of the flywheel enclosure (1) as a safety response to an alarm or fault condition, (2) to limit precession travel beyond the functional maximum, or (3) as the standard means to shut down the gyroscopic roll stabilizer.

[0013] Accordingly, it is an object of the disclosure not to encompass within the disclosure any previously known product, process of making the product, method of using the product, or method of treatment such that Applicants reserve the right and hereby disclose a disclaimer of any previously known product, process, or method. It is further noted that the disclosure does not intend to encompass within the scope of the disclosure any product, process, or making of the product or method of using the product which does not meet the written description and enablement requirements of the USPTO (35 U.S.C. § 112, first paragraph) or the EPO (Article 83 of the EPC), such that Applicants reserve the right and hereby disclose a disclaimer of any previously described product, process of making the product, or method of using the product disclosed herein.

[0014] It is noted that in the present disclosure and particularly in the claims and/or paragraphs, terms such as “comprises,” “comprised,” “comprising” and the like can have the meaning attributed to them in U.S. patent law; for example, they can mean “includes,” “included,” “including,” and the like; and that terms such as “consisting essentially of” and “consists essentially of” have the meaning ascribed to them in U.S. patent law; for example, they allow for elements not explicitly recited, but exclude elements that are found in the prior art or that affect a basic or novel characteristic of the invention.

[0015] These and other embodiments are disclosed or are obvious from and encompassed by the following Brief Description of the Drawings and Detailed Description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. **1** demonstrates an embodiment of the present disclosure, wherein a pawl is mechanically linked to a solenoid plunger and can be activated when power is terminated.

[0017] FIG. **2** depicts a side view of the pawl and solenoid plunger of FIG. **1**.

[0018] FIG. **3** depicts a perspective view of the pawl and solenoid plunger of FIG. **1**.

[0019] FIG. **4** depicts a second perspective view of the pawl and solenoid system attached to the mounting bracket by a pawl compliance shaft.

[0020] FIG. **5** depicts a third perspective view of the pawl and solenoid system attached to the mounting bracket by a pawl compliance shaft.

[0021] FIG. **6** depicts an exploded view of the components of the pawl, solenoid system, and connecting hardware.
[0022] FIG. **7** depicts an embodiment of the pawl in a locked configuration.
[0023] FIG. **8** depicts a side view of the pawl detached from the solenoid system.
[0024] FIG. **9** depicts a second side view of the pawl detached from the solenoid system.
[0025] FIG. **10** depicts a front view of the pawl detached from the solenoid system.
[0026] FIG. **11** depicts a perspective view of a gyroscopic roll stabilizer when the flywheel enclosure is in a rested position.
[0027] FIG. **12** depicts a side view of the gyroscopic roll stabilizer of FIG. **11**.
[0028] FIG. **13** depicts a side view of the gyroscopic roll stabilizer of FIG. **11** with the side rail and gimbal shaft cover removed.
[0029] FIG. **14** depicts an exploded view of the flywheel enclosure and flywheel from the gyroscopic roll stabilizer of FIG. **11**.
[0030] FIG. **15** depicts a perspective view of a gyroscopic roll stabilizer when the flywheel enclosure is rotated in a forward position.
[0031] FIG. **16** depicts a side view of the gyroscopic roll stabilizer of FIG. **15**.
[0032] FIG. **17** depicts a side view of the gyroscopic roll stabilizer of FIG. **15** with the side rail and gimbal shaft cover removed and the parking pawl engaged with the gimbal shaft.
[0033] FIG. **18** depicts a perspective view of a gyroscopic roll stabilizer when the flywheel enclosure is rotated in a rear position.
[0034] FIG. **19** depicts a side view of the gyroscopic roll stabilizer of FIG. **18**.
[0035] FIG. **20** depicts a side view of the gyroscopic roll stabilizer of FIG. **18** with the side rail and gimbal shaft cover removed and the parking pawl engaged with the gimbal shaft.

DETAILED DESCRIPTION

[0036] The following detailed description, given by way of example, but not intended to limit the invention solely to the specific embodiments described, may best be understood in conjunction with the accompanying drawings.

[0037] The present disclosure relates generally to an assembly, a system and a method **100** to safely stop a flywheel enclosure **202** in a gyroscopic roll stabilizer **200** from precessing in an uncontrolled manner in the event of a braking or electrical failure. In an embodiment, the flywheel enclosure **202** houses a flywheel **204** which is disposed in a neutral orientation wherein the axis of rotation of the flywheel **204** is generally parallel to either of the vertical or horizontal axis of the vessel that the gyroscopic roll stabilizer **200** is affixed to. The flywheel **204** rotates within the flywheel enclosure **202** causing the gyroscopic roll stabilizer **200** to rotate. In an embodiment, the gyroscopic roll stabilizer **200** may be affixed to a vessel by fastening the side rail **206** of the gyroscopic roll stabilizer **200** to the vessel.

[0038] The assembly, method, and system to prevent precession of the gyroscopic roll stabilizer **200** may comprise a pawl **102**, a gimbal shaft **110**, apertures **126** capable of receiving the pawl **102**, and an energized solenoid **118** (having a solenoid housing **136**) and solenoid plunger **120** capable of exerting a force on one end of the pawl **102** during an electric, hydraulic, or mechanical failure event. The assembly or system is generally depicted by FIG. **1** (the “Parking Pawl System”). The flywheel enclosure **202** is fastened to rotating surface **130** which follows the rotation of the flywheel enclosure **202**. The gimbal shaft **110** is also fastened to the rotating surface **130** and the movement of the flywheel enclosure **202** and rotating surface **130** cause the gimbal shaft **110** to rotate.

[0039] In an embodiment, a gimbal shaft cover **208** is disposed to protect the gimbal shaft **110** from external elements.

The Pawl

[0040] FIGS. **1** and **13** depict a pawl **102** in a disengaged configuration in relation to gimbal shaft **110**. The pawl **102** comprises a first end with a latching pawl tooth **134**, and a second end with a solenoid plunger connector **132**. The first end of the pawl with a latching pawl tooth **134** further comprises a pawl front latching pad **104**, a pawl back latching pad **106**, and a pawl end stop **108**. If the gyroscopic roll stabilizer **200** is operating properly, then the pawl **102** will remain disengaged from the gimbal shaft **110**, and an operator can turn off the gyroscopic roll stabilizer **200** by means of a standard brake system.

[0041] The latching pawl tooth **134** of the pawl **102** may comprise a variety of shapes and configurations and is configured to correspond to the shape of the apertures **126** disposed along edges of the gimbal shaft **110** to optimize the locking action of the Parking Pawl System.

[0042] FIGS. **8-10** depict one preferred embodiment of the pawl **102**, the latching pawl tooth **134**, and the pawl end stop **108**. In an embodiment, the linear distance between the center of the hole **160** in the pawl **102** and a first side of latching pawl tooth **134** is approximately between 3 to 5 inches. In a preferred embodiment, the linear distance between the center of the hole **160** and the first side of the latching pawl tooth **134** is approximately between 3.9 to 4.060 inches.

[0043] In an embodiment, the linear distance between the hole **160** and the second side of the latching pawl tooth **134** is approximately between 2 to 4 inches. In a preferred embodiment, the linear distance between the center of the hole **160** and the second side of the latching pawl tooth **134** is approximately between 3.090 to 3.350 inches.

[0044] In an embodiment, the angle between a plane rising vertically from the center of the hole **160** and the plane extending from the edge of the front latching pad **104** is approximately in the range of 16 to 18 degrees. In a preferred embodiment, the angle between a plane rising vertically from the center of the hole **160** and the plane extending from the edge of the front latching pad **104** is approximately 17 degrees.

[0045] In another embodiment, the angle between a plane rising vertically from the center of the hole **160** and the plane extending from the edge of the back latching pad **106** is approximately 16.8 to 18.8 degrees. In a preferred embodiment, the angle between the plane rising vertically from the center of the hole **160** and the plane extending from the edge of the back latching pad **106** is approximately 17.81 degrees.

[0046] In an embodiment, the distance between the plane formed by the pawl end stop **108** and a parallel plane extending from the center of the hole **160** is approximately between 2 to 4 inches. In the preferred embodiment, the distance between these planes is approximately 3.008 inches. The angle formed by a plane extending from the pawl end stop **108** and a plane extending horizontally from the center of the hole **160** is approximately between 21 to 23 degrees. In the preferred embodiment, the angle between these planes is approximately 22 degrees.

[0047] In an embodiment, the distance from the bottom side of the pawl **102** to the topmost portion of the latching pawl tooth **134** is approximately between 1.5 to 2.5 inches. In the preferred embodiment, the distance is approximately 1.821 inches.

[0048] In one preferred embodiment of the present disclosure, the latching pawl tooth **134** and apertures **126** are configured to lock together while the gimbal shaft **110** is mid-rotation to bring the rotation of the gimbal shaft **110** to an immediate stop. The latching pawl tooth **134** is designed and operationally tested under full load to achieve the lock action when the gimbal shaft **110** is rotating at significant angular velocity. The latching pawl tooth **134** is configured to remain engaged with the apertures **126** after an abrupt stop or brake.

[0049] In one embodiment of the present disclosure, the latching pawl tooth **134** and apertures **126** may lock together while the gimbal shaft **110** is precessing at a speed of up to 650 degrees per second. However, larger or smaller embodiments of the present disclosure may be developed to achieve the same objectives, at a different scale, to achieve the intended functionality. For instance, in smaller alternative embodiments, the latching pawl tooth **134** and apertures **126** are configured to lock together at lower precessing speeds, and in larger alternative embodiments, the latching pawl tooth **134** and apertures **126** are configured to lock together at high precessing speeds.

Braking System

[0050] In an embodiment of the present invention, the Parking Pawl System acts as a backup to a standard brake system for the flywheel enclosure **202**. The braking system may be any applicable or desirable braking system that achieves the intended purpose as would be understood by one of ordinary skill in the art. FIGS. **1** and **11-20** depict elements of a standard braking system which may include a brake actuator (or "braking actuator") **124** and brake actuator rod (or "actuator rod") **122**. As shown, the brake actuator rod **122** is mechanically attached to the gimbal shaft **110** and capable of slowing the rotation of the flywheel enclosure **202** when activated. When the braking system is not activated, the solenoid **118** remains energized by an electricity source, such as a battery, and does not engage the solenoid plunger **120** or exert a force on the pawl **102**. FIGS. **2-5** depict the solenoid plunger **120** connected to the pawl **102** by means of a connection rod **158**. The solenoid plunger **120** is connected at one end to the pawl **102** and connected at the other end to a solenoid housing **136**.

[0051] FIG. **6** depicts an exploded view of the Parking Pawl System **100**. As shown in FIG. **6**, the pawl **102** is operatively connected to a solenoid plunger **120** and the solenoid plunger **120** is detachably connected to the solenoid housing **136**.

[0052] The solenoid plunger **120** is connected to pawl **102** via a connecting rod **158** and two roll pins **156**. A first end of the connecting rod **158** is positioned in a groove in an end of the solenoid plunger **120** and a second end of the connecting rod **158** is positioned in a groove in an end of the pawl **102**. A first roll pin **156** extends through a series of holes proximate to the groove in the solenoid plunger **120** such that a first roll pin **156** extends through the plunger and the connecting rod **158**. A second roll pin **156** extends through the pawl **102** (at the solenoid plunger connector **132**) and the connecting rod **158**. FIGS. **3-4** depict the roll pins **156** and connecting rod **158** configured to connect the solenoid plunger **120** to the pawl **102**.

[0053] A spring **154**, washer **146**, nut **152**, and mounting bracket **148** allow the solenoid plunger **120** to exert a force on one end of the pawl **102** in the event that solenoid **118** and the solenoid plunger **120** are de-energized, such as in the event of an electrical failure. When energized, the solenoid **118** retracts the solenoid plunger **120** into a non-extended disposition, compressing the spring **154** between the solenoid **118** and pawl **102**. When the solenoid plunger is de-energized, the spring **154** expands and applies a force on the pawl **102** to cause rotation of the pawl **102**.

[0054] A mounting bracket **148** may be affixed to any component of an aquatic vessel as may be necessary, and may be affixed with a plurality of screws **150** and washers **144**, as depicted in FIG. **6**. FIG. **6** further depicts components which allow the pawl **102** to rotate when needed. A hole **160** in the pawl **102** may receive a pawl pivot bushing **116**. The pawl pivot bushing **116** may receive a pawl compliance sleeve **112**. The pawl compliance sleeve **112** contains a plurality of O-rings **138**. The O-rings **138** surround a 1-inch compliant latch mechanism shaft **114**.

[0055] In an embodiment, an additional O-ring **138** may be situated on the opposite side of the 1-inch compliant latch mechanism shaft **114**. The O-ring **138** on the opposite side of the 1-inch compliant latch mechanism shaft **114** is positioned substantially around a washer wedge lock **140**. A fastener socket **142** extends through the washer wedge lock **140** and into the 1-inch compliant latch mechanism shaft **114** and is capable of receiving a fastener **146**.

[0056] In an embodiment, a fastener **146** may be inserted through a washer **144** on one side of the pawl **102** and through the hole **160** to connect with the fastener socket **142** on the opposite side of the pawl **102** to ensure that the components depicted in FIG. **6** remain connected and operational.

[0057] The components depicted in FIG. **6** permit the pawl **102** to rotate around the axis formed by the fastener **146** and the fastener socket **142** such that when a force is exerted on one end of the pawl **102** by the solenoid plunger **120**, the other end of the pawl **102** moves to engage with the apertures **126** in the gimbal shaft **110** to stop unwanted precession of the flywheel enclosure **202**.

[0058] In the event that the standard braking system successfully stops the rotation of the flywheel enclosure **202**, the Parking Pawl System **100** may be activated to act as a backup system to further keep the flywheel enclosure **202** from rotating.

Braking System Failure

[0059] In the event of a mechanical, electrical, or hydraulic failure in the braking system or if the gyroscopic roll stabilizer **200** experiences a loss of power, the braking actuator **124** may be unable to actuate the actuator rod **122** and thus may be unable to slow the motion of the flywheel enclosure **202** and the Parking Pawl System **100** will be activated. FIGS. **7**, **17**, and **20** depict the pawl **102** in locked configurations wherein the latching pawl tooth **134** is engaged with an aperture **126** in the gimbal shaft **110**. The Parking Pawl System **100** may also be activated if a safety alarm or a fault condition is detected in any operations system or if the flywheel enclosure **202** precession travel is beyond its functional maximum. In such events, the standard braking system may be unable to stop uncontrolled flywheel enclosure precession, which can, if unbraked, reach angular velocities of up to 600 degrees/second and cause serious damage to the aquatic vessel or bodily harm to its occupants.

[0060] For example, the braking actuator **124** and actuator rod **122** may be a hydraulic braking system. Should the hydraulic braking system lose all hydraulic pressure, or if the components themselves actually break, the hydraulic braking system will fail and lose control of the rotation of the flywheel enclosure **202**.

[0061] The solenoid **118** is part of the Parking Pawl System **100** and is a redundant and wholly separate braking system from the braking actuator **124** and actuator rod **122**.

[0062] In the event of a mechanical failure or electrical failure of the braking system, the solenoid **118** will engage the solenoid plunger **120**. When energized, the solenoid **118** retracts the solenoid plunger **120** into a non-extended disposition. The solenoid **118** remains energized when the system is operational and holds back the solenoid plunger **120** from moving to engage with the pawl **102**. If the gyroscopic stabilizer **200** loses power, or if another fault triggers an alarm, the solenoid **118** will de-energize and the spring **154** will cause the solenoid plunger **120** to extend, thereby engaging the motion of the pawl **102**. When the solenoid **118** is de-energized, the solenoid plunger **120** will be engaged. When engaged, the solenoid plunger **120** exerts a force on one end of the pawl **102**. The solenoid plunger **120** is mechanically coupled to one end of the pawl **102** at the solenoid plunger connector **132**. When the solenoid plunger **120** exerts a force on the pawl **102**, the pawl **102** rotates around a rotation point.

[0063] In an embodiment, the pawl rotation point includes a pawl pivot bushing **116**, a pawl compliance shaft **114**, and a pawl compliance sleeve **112**. Together, the pawl pivot bushing **116**, the pawl compliance shaft **114**, and the pawl compliance sleeve **112** allow the pawl **102** to rotate in such a way that when one end of the pawl **102** is moved by the solenoid plunger **120**, the other end of the pawl **102** (the latching pawl tooth **134**) is able to move into an engaged configuration with the apertures **126** of the gimbal shaft **110**, thereby locking the flywheel enclosure **202** from continuing to exert a rotation force.

[0064] Whereas one end of the pawl **102** is coupled to the solenoid plunger **120**, the other end of the pawl **102** comprises a latching pawl tooth **134**. The latching pawl tooth **134** can be inserted into one of multiple apertures **126** in the gimbal shaft **110**. Once the latching pawl tooth **134** is engaged in an aperture **126** of the gimbal shaft **110**, the stabilization system will brake regardless of whether there is power supplied to the system or not. Because the solenoid **118** activates the solenoid plunger **120** when there is a loss of power, thereby engaging the pawl **102** to brake the system, any rotation of the flywheel enclosure **202** will be controlled, even if other systems on a vessel have failed. This is preferred to other emergency braking systems, such as hydraulic braking systems, which are more complex, require electricity to operate, and may not remain engaged or locked in the event of power loss.

Apertures and Pawl Latching End

[0065] The latching pawl tooth **134** is configured to fit into at least one corresponding aperture in the gimbal shaft **110**. Two apertures **126** are positioned within the gimbal shaft **110**. In the preferred embodiment, the two apertures **126** are positioned at the maximal rotational angle of the gimbal shaft **110**. This allows the pawl **102** to lock the movement to the gimbal shaft **110** immediately once the pawl latching pad engages with the aperture **126**. Alternative embodiments may comprise greater than or less than two apertures **126** which are positioned within the gimbal shaft **110** at any number of angles or desirable positions.

[0066] The shape of the pawl latching end shown in FIG. **1** is shown as an example, but the latching pawl tooth **134** may be any geometric shape so as to maximize locking ability within the apertures; likewise, the corresponding apertures **126** may be any shape capable of receiving the latching pawl tooth **134**. The aperture **126** and the latching pawl tooth **134** are preferably shaped to maximize locking power and stability even when the gimbal shaft is rotating with significant angular velocity. Preferably, the latching pawl tooth **134** of the pawl **102** is configured such that once it is locked into a corresponding aperture **126**, the latching pawl tooth **134** cannot disengage accidentally.

[0067] FIGS. **15-17** depict the gyroscopic roll stabilizer **200** rotated in a forward position with the pawl **102** activated to prevent rotation of the gyroscopic roll stabilizer **202**. When the gyroscopic roll stabilizer **200** is rotated in the forward position, the latching pawl tooth **134** is disposed within an aperture **126** of the gimbal shaft **110** such that the back latching pad **106** is disposed in contact with an internal wall of the aperture **126**. In an embodiment, a flat edge of the back latching pad **106** is flush with an internal wall of the aperture **126**. Similarly, at least a portion of the pawl end stop **108** is disposed in contact with another internal wall of the aperture **126**. In an embodiment, a flat edge of pawl end stop **108** is flush with a portion of an internal wall of the aperture **126**.

[0068] In the embodiment shown in FIG. **17**, the latching pawl tooth **134** is disposed within the aperture **126** such that it prevents the gimbal shaft **110** from rotating back to a rested position (as seen in FIGS. **11-14**) or a rearward position. The back latching pad **106** applies force on the wall of the aperture which prevents the gimbal shaft **110** from rotating back to its rested position (or a rearward position). If the gimbal shaft **110** continued to rotate in a forward direction, the front latching pad **104** would come into contact with an internal wall of the aperture **126** and prevent further forward rotation of the gimbal shaft **110**.

[0069] FIGS. **18-20** depict the gyroscopic roll stabilizer **200** rotated in a rearward position with the pawl **102** activated to prevent rotation of the gyroscopic roll stabilizer **202**. When the gyroscopic roll stabilizer **200** is rotated in the rearward position, the latching pawl tooth **134** is disposed within an aperture **126** of the gimbal shaft **110** such that the front latching pad **104** is disposed in contact with an internal wall of the aperture **126**. In an embodiment, a flat edge of the front latching pad **104** is flush with an internal wall of the aperture **126**. Similarly, at least a portion of the pawl end stop **108** is disposed in contact with another internal wall of the aperture **126**. In an embodiment, a flat edge of pawl end stop **108** is flush with a portion of an internal wall of the aperture **126**.

[0070] In the embodiment shown in FIG. **20**, the latching pawl tooth **134** is disposed within the aperture such that it prevents the gimbal shaft **110** from rotating back to a rested position (as seen in FIGS. **11-14**) or a forward position. The front latching pad **104** applies a force on the wall of the aperture **126** which prevents the gimbal shaft **110** from rotating back to its rested position (or a forward position). If the gimbal shaft **110** continued to rotate in a rearward direction, the back latching pad **106** would come into contact with an internal wall of the aperture **126** and prevent further rearward rotation of the gimbal shaft **110**.

[0071] In an embodiment, and as depicted in FIGS. **11-20**, the apertures **126** of the gimbal shaft **110** comprise 3 internal walls which generally correspond with the front latching pad **104**, back latching pad **106**, and pawl end stop **108**. A first aperture **126** is used to stop the gimbal shaft in a forward position, and a second aperture **126** is used to stop the gimbal shaft in a rearward position. In alternative embodiments, the gimbal shaft **110** may comprise a single aperture **126** or more than two apertures **126**.

Catastrophic Failure Test Results

[0072] It has been observed that during a catastrophic failure, the flywheel enclosure **202** is capable of rotating at a rate of approximately 600 degrees/second. This angular velocity can take hours to slow down and stop if unrestrained by the brake system or the pawl **102**. An angular velocity of this magnitude is also capable of causing serious damage to a vessel and causing injury to vessel occupants.

[0073] The pawl **102** disclosed herein has been tested under full load conditions and was successfully able to stop uncontrolled rotation of the flywheel enclosure **202**. In particular, the pawl **102** was observed to engage with the aperture and lock the motion of the flywheel enclosure **202** at an angular velocity of up to 600 degrees/second (the maximal observed rate of rotation in an uncontrolled system). In an embodiment, the pawl **102** may engage with the aperture and lock the motion of the flywheel enclosure **202** at an angular velocity of greater than 600 degrees/second.

Pawl Misalignment Test Results

[0074] Under normal conditions, the pawl **102** and the gimbal shaft **110** face are preferably coplanar and parallel to one another. This encourages secure locking of the latching pawl tooth **134** of the pawl **102** into the apertures **126** positioned on the gimbal shaft **110**. However, it is possible, during a failure or otherwise, for the pawl **102** and the gimbal shaft **110** face to be or become misaligned.

[0075] Misalignment of the pawl **102** and the gimbal shaft **110** may naturally occur through use of the device. In a preferred embodiment, the Parking Pawl System **100** uses fasteners with tight tolerances to fasten components and minimize misalignment.

[0076] It was observed that the pawl **102** is capable of successfully locking with the gimbal shaft **110** even when misaligned. In particular, it was observed that the pawl **102** was capable of forming a locking engagement with the apertures **126** in the gimbal shaft **110** with a frame misalignment of 1.5 degrees between the pawl **102** and the gimbal shaft **110** and no longer co-planar nor parallel to one another.

[0077] The Parking Pawl System **100** operates most effectively when the gimbal shaft **110** and the pawl **102** are misaligned by 1.0 degree or less. However, the pawl **102** is designed to cease the rotation of the flywheel enclosure **202** even in the event that the vertical plane of the gimbal shaft **110** and the vertical plane of the pawl **102** are misaligned by up to 1.5 degrees.

Multi-Directional Testing

[0078] In the course of operating within a gyroscopic roll stabilizer **200**, angular momentum moves the gimbal shaft **110** in multiple directions. The Parking Pawl System **100** is capable of stopping the movement of the gimbal shaft **110** regardless of its current direction of movement. Specifically, the pawl **102** is designed to successfully achieve a locking configuration with the gimbal shaft **110** by engaging the latching pawl tooth **134** into a corresponding aperture **126** while the gimbal shaft **110** is rotating with significant angular velocity in either direction of movement. This ensures that the pawl **102** will be able to lock to the gimbal shaft **110** immediately, regardless of where the gimbal shaft **110** is positioned in an arc of movement when a braking failure occurs.

[0079] Although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions, and alterations can be made herein without departing from the spirit and scope of the invention as defined in the appended claims. Having thus described in detail preferred embodiments of the present invention, it is to be understood that the invention defined by the above paragraphs is not to be limited to particular details set forth in the above description, as many apparent variations thereof are possible without departing from the spirit or scope of the present invention.

Claims

1. An emergency brake to prevent precession of a gyroscopic flywheel enclosure in the event of a braking failure, the emergency brake comprising: a solenoid plunger having an internal portion disposed within a cavity of a solenoid and an external portion extending downward from the cavity

of the solenoid, the solenoid plunger oriented to move in a linear path along a first axis; a pawl having a first end, a second end, and a through-hole disposed therebetween, the first end of the pawl rotatably connected to the solenoid plunger; a pivot disposed within the through-hole of the pawl, the pivot oriented to rotate the pawl around a second axis; a gimbal shaft having a perimeter wall and a plurality of apertures disposed along the perimeter wall, the gimbal shaft oriented to oscillate about a third axis in parallel with the second axis; wherein a latching pawl tooth extends from the second end of the pawl and is configured to engage with one of the plurality of apertures disposed along the perimeter wall of the gimbal shaft; and wherein the solenoid, pawl, and gimbal shaft are operatively connected to a gyroscopic roll stabilizer and the gyroscopic roll stabilizer oscillates the gimbal shaft about the third axis.
2. The emergency brake of claim 1, wherein the emergency brake system moves between a disengaged position and an engaged position.
3. The emergency brake of claim 2, wherein when the system is in the disengaged position, the latching pawl tooth is spaced apart from the gimbal shaft and the gimbal shaft freely oscillates about the third axis.
4. The emergency brake of claim 2, wherein the solenoid plunger moves in a downward direction away from the solenoid to rotate the pawl from a disengaged position to an engaged position, wherein the solenoid plunger applies downward pressure to the first end of the pawl causing the second end of the pawl to rotate upward about the second axis; and wherein the latching pawl tooth interlocks with one of the plurality of apertures of the gimbal shaft when the pawl is in the engaged position.
5. The emergency brake of claim 2, wherein the latching pawl tooth extends perpendicular from the second end of the pawl and the latching pawl tooth comprises: a front latching pad forming a first edge of the latching pawl tooth, a back latching pad forming an opposing second edge of the latching pawl tooth, and a pawl end stop forming a third edge and connecting the front and back latching pad.
6. The emergency brake of claim 2, wherein when the system is in the engaged position, the latching pawl tooth is disposed within the apertures of the gimbal shaft and the latching pawl tooth prevents the gimbal shaft from freely oscillating about the third axis; wherein an internal edge of one of the plurality of apertures of the gimbal shaft is in interlocking exchange with the first edge of the latching pawl tooth.
7. An emergency brake system for preventing precession of a gyroscopic flywheel enclosure comprising: a gimbal shaft rotatable about a first axis and having a plurality of apertures disposed along a perimeter edge of the gimbal shaft; a pawl having a first end and a second end, wherein the pawl is rotatable around a second axis disposed between the first end and second end; wherein the pawl and gimbal shaft are disposed within the same plane; wherein a latching pawl tooth extends from the first end of the pawl, the latching pawl tooth being configured to engage with one of the plurality of apertures disposed along the edge of the gimbal shaft; and a solenoid plunger partially disposed within a cavity of a solenoid and connected to the second end of the pawl.
8. The emergency brake system of claim 7, further comprising a power source electronically connected to the solenoid, such that when the power source is active the solenoid plunger is retracted within the cavity of the solenoid.
9. The emergency brake system of claim 8, further comprising a spring compressed between the second end of the pawl and the cavity of the solenoid, wherein when the power source is deactivated: the spring de-compresses and applies a force to the solenoid to extend the solenoid plunger from the cavity of the solenoid; and the solenoid plunger pushes the second end of the pawl downward to rotate the first end of the pawl about the second axis.
10. The emergency brake system of claim 9, wherein, when the first end of the pawl is rotated about the second axis, the latching pawl tooth engages with one of the plurality of apertures disposed along the perimeter edge of the gimbal shaft.

11. A method of braking a gyroscopic roll stabilizer where the gyroscopic roll stabilizer comprises a flywheel rotating within a flywheel enclosure causing oscillation of a gimbal shaft, the method comprising the steps of: de-energizing a solenoid to extend a solenoid plunger along a first axis away from a solenoid housing, wherein the solenoid plunger is disposed within a cavity of the solenoid housing; applying a downward force to a first end of a pawl with the solenoid plunger, wherein the solenoid plunger is connected to the first end of the pawl; rotating a second end of the pawl about a second axis, the second end of the pawl comprising a latching pawl tooth; and interlocking the latching pawl tooth within an aperture disposed within an edge of a perimeter of the gimbal shaft to prevent oscillation of the gimbal shaft.
12. The method of claim 11, wherein the latching pawl tooth extends generally perpendicular from the second end of the pawl.
13. The method of claim 11, wherein prior to de-energizing the solenoid, the flywheel enclosure is oscillating at a speed of greater than 600 degrees per second.
14. The method of claim 11, wherein prior to de-energizing the solenoid, a first braking system fails to prevent uncontrollable precession of the flywheel enclosure.
15. The method of claim 11, wherein after interlocking the latching pawl tooth within the aperture of the gimbal shaft, the method comprises the additional steps of: re-energizing the solenoid to retract the solenoid plunger along the first axis into the cavity of the solenoid housing; applying an upward force to the first end of the pawl with the solenoid plunger as the solenoid plunger retracts into the cavity of the solenoid housing; and rotating the second end of the pawl about the second axis to remove the latching pawl tooth from the aperture of the gimbal shaft.